Advanced Labs - Theme D: Optoelectronics

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1 Part 1

1.1 Emission Spectrum Graphs

The emission spectrum for the red-light LED at room temperature is shown in Fig 1.

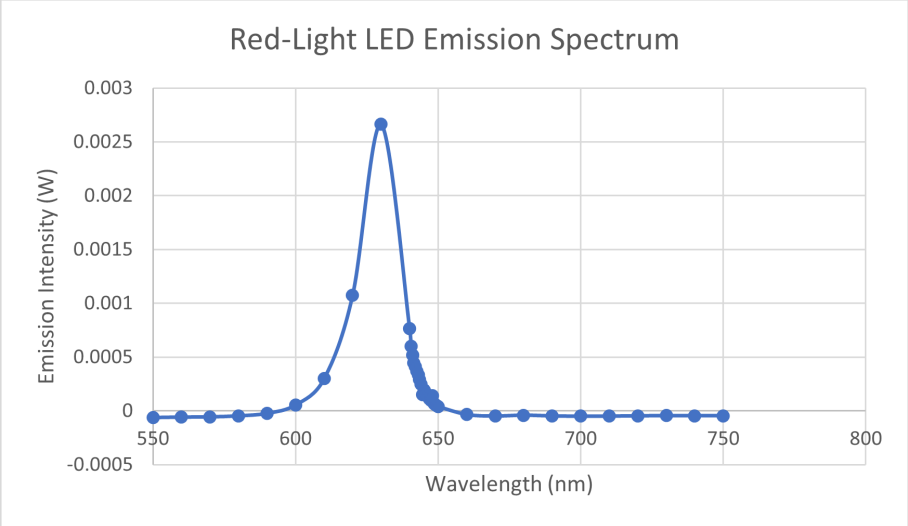


Figure 1: Emission Spectrum for the red-light LED at room temperature

The emission spectrum for the blue-light LED at $60°C$ and $3°C$ is shown in Figures 2 and 3 respectively.

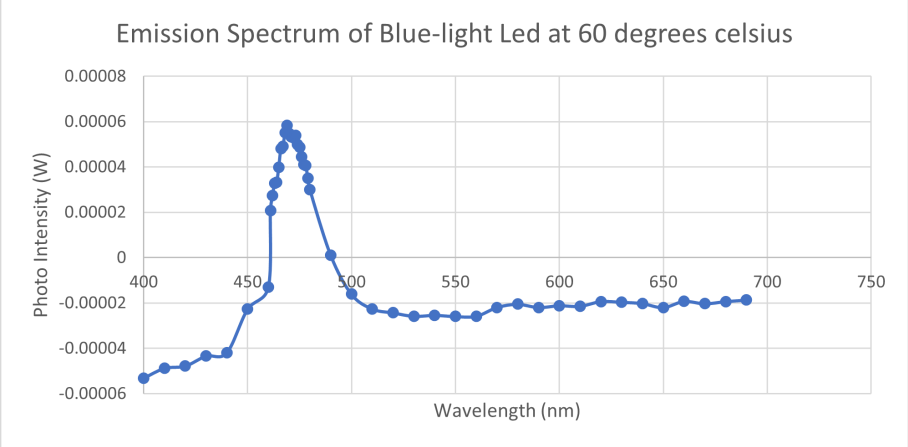


Figure 2: Emission Spectrum for the blue-light LED at $60°C$

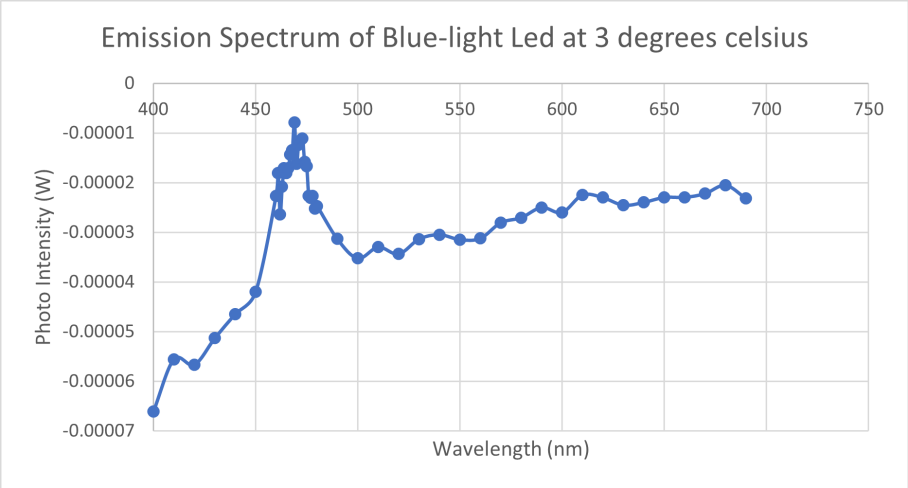


Figure 3: Emission Spectrum for the blue-light LED at $3°C$

The emission spectrum for yellow-light LED is shown in Fig 4.

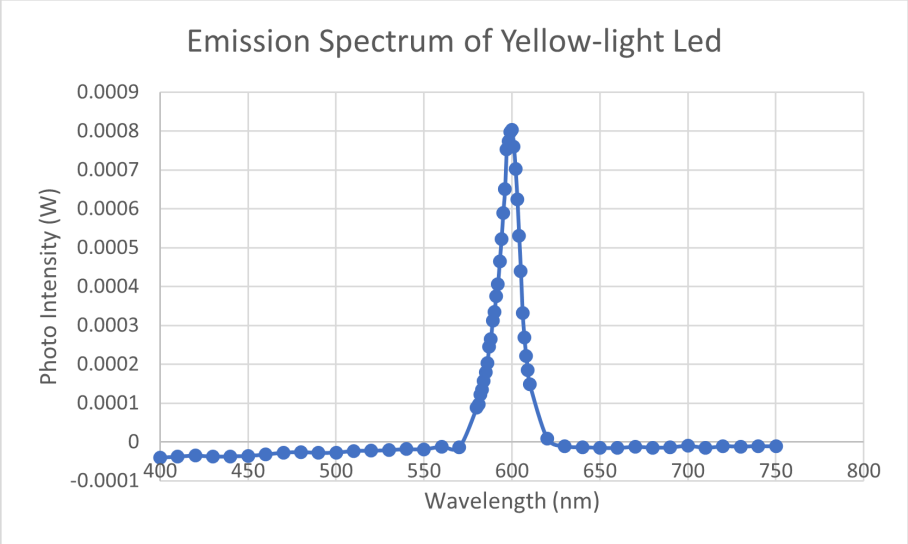


Figure 4: Emission Spectrum for the yellow-light LED at room temperature

The emission spectrum for white-light LED is shown in Fig 5.

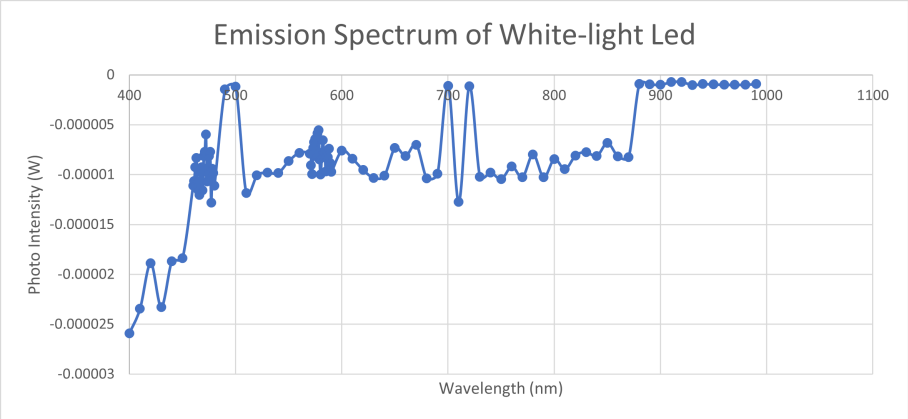


Figure 5: Emission Spectrum for the White-light LED at room temperature

1.2 Peak Wavelengths

To find the peak Wavelengths we just identify the highest points on the graph, except for blue light which is found by identifying the highest point and averaging it with the point at which the emission intensity becomes half. Note that white-light LED has 3 peaks. The peaks are shown in Table 1.

LED	Peak Wavelength	Emission Intensity
Red	630	0.002663415
Blue (3)	473	-9.49074E-06
Blue (60)	476.5	4.19437E-05
Yellow	600	0.000803252
White -1	500	1.18774E-06
White -2	700	1.08796E-06
White -3	720	1.13379E-06

Table 1: Peak Wavelengths

1.3 White light production

The White-light LED has three main peaks in its spectrum, and these peaks are at about the same wavelengths as those in Red, Yellow, and Blue LEDs. We know that white color is really a mix of all colors. So, with this in mind, the white light from a White-light LED is made by mixing the lights from Red, Blue, and Yellow LEDs. If we change how bright each of these colored LEDs shines, we can make different kinds of white colors.

1.4 Filter change

We switch the filter when measuring the White-light LED's emission spectrum because the straight white light from the LED includes other wavelengths besides our target wavelength. These extra wavelengths can pass through the diffraction grating without any trouble. So, to get rid of these unwanted wavelengths, we use a special filter (like the R-64 filter in our experiment) that blocks them. It's worth mentioning that this problem mainly occurs when we measure wavelengths higher than about 690 nm.

1.5 blue-light LED changes

According to the additional details found in Supplemental Material 1 from our textbook, the reasoning behind the diminished peak intensity of the Blue-light LED at higher temperatures becomes clear. While the rate of radiative recombination (the process that yields a photon) stays the same when the temperature goes up, the rate of non-radiative recombination (which leads to the excitation of another electron instead of photon emission) sees an increase. The supplemental material also clarifies that the radiative recombination ratio, Rrad can be expressed by the formula below. This ratio is directly proportional to the light output of the LED:

$$R_{\text{rad}} = \frac{1}{\frac{1}{\tau_{\text{rad}}} + \frac{1}{\tau_{\text{non-rad}}}}$$

Considering this, it's evident that as the temperature rises, R_{rad} diminishes, and since it's directly proportional to light output, the LED's emission intensity correspondingly declines with an increase in temperature.

1.6 Red-light LED changes

When we significantly increase the current flowing into the Red-light LED, the light intensity in the area where the light is produced goes above the threshold level. This means there are a lot more carriers and electrons ready to come together and create light particles, or photons. As a result, the LED starts to emit a lot more light.

2 Part 2

2.1 Xe lamp Graph

The emission spectrum of the Xe lamp is shown in Fig 6.

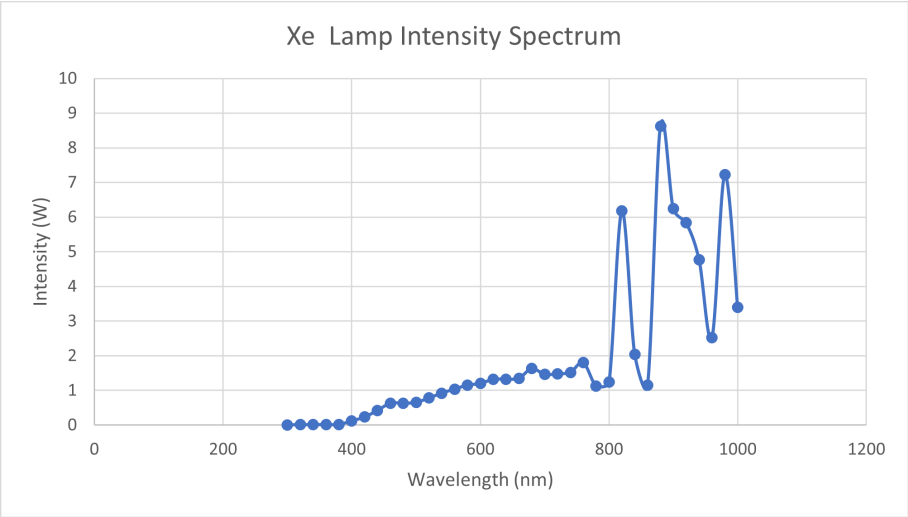


Figure 6: emission spectrum of the Xe lamp

2.2 Solar Cells Spectra

The photoresponse spectrum of solar cell A is shown in Fig 7

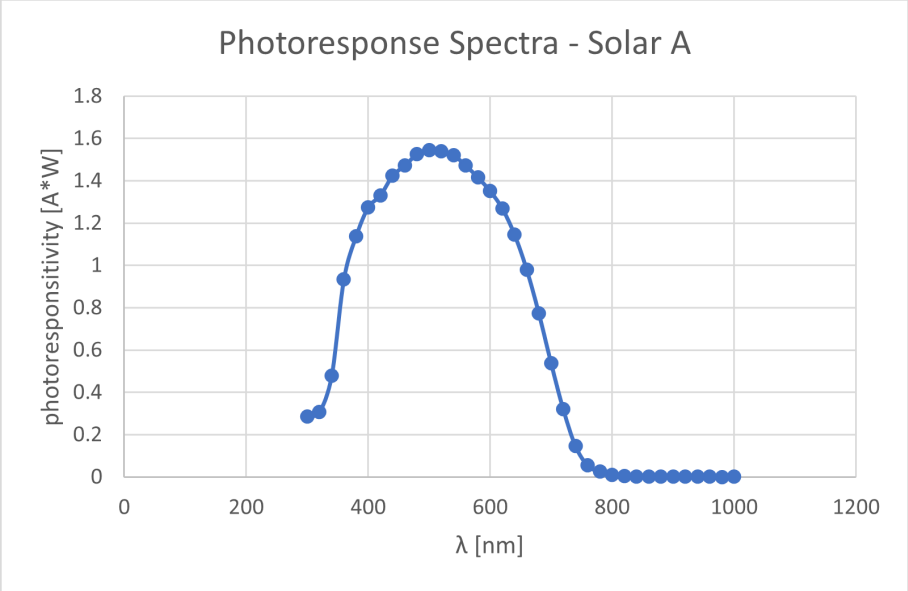


Figure 7: photoresponse spectrum of solar cell A

The photoresponse spectrum of solar cell B is shown in Fig 8

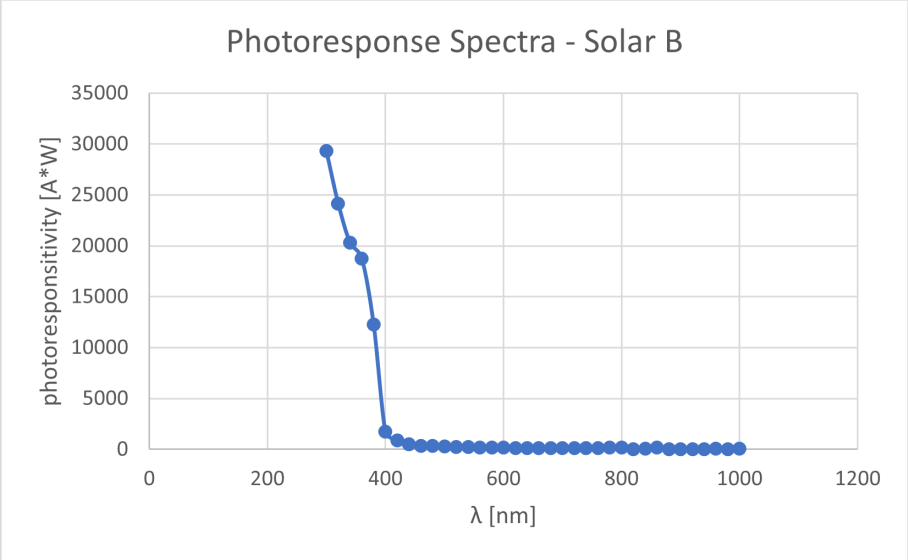


Figure 8: photoresponse spectrum of solar cell B

The photoresponse spectrum of solar cell C is shown in Fig 9

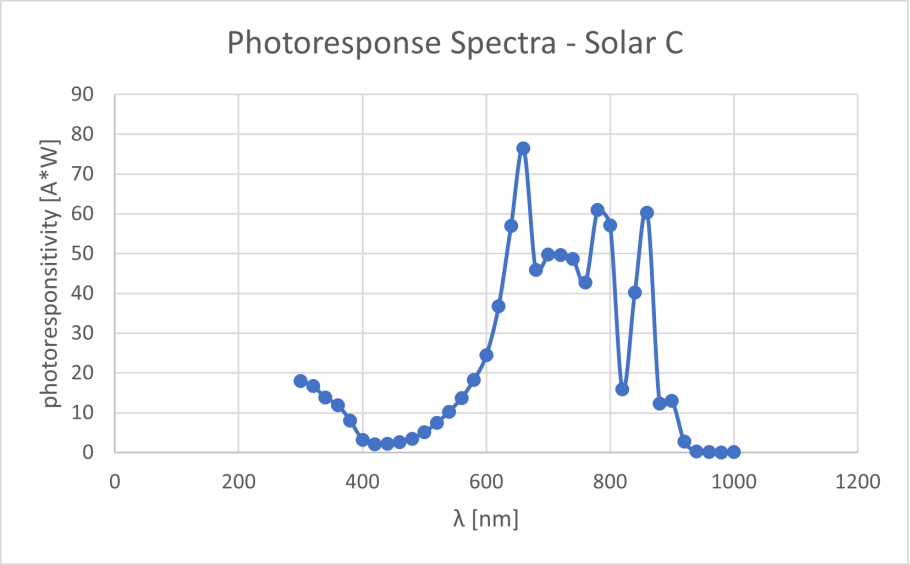


Figure 9: photoresponse spectrum of solar cell C

2.3 Solar Cells Identification

From the results of our experiments, we've determined the types of solar cells:

- Solar Cell A is made of Amorphous Silicon (a-Si).
- Solar Cell B is made of Crystalline Silicon (c-Si).
- Solar Cell C is a Tandem Structure with Gallium Arsenide (GaAs) compounds.

We concluded this from their Quantum Efficiency (QE) spectra. Solar Cell A showed one peak followed by a gradual drop in QE, indicating ongoing recombination, typical for amorphous silicon. In contrast, Solar Cell B had an initial peak and then a sharp fall in QE, stabilizing afterward, which is characteristic of crystalline silicon. Finally, Solar Cell C displayed two peaks in its QE spectrum, aligning with the expected behavior of tandem-structure cells.

2.4 Quantum Efficiency

The relationship between quantum efficiency and photoresponsivity is found to be:

$$\eta = R \times \hbar\overline{\omega} \times 100\% \tag{1}$$

We can derive this equation by:

$$\eta = \frac{Iphv}{qP_{in}} \tag{2}$$

$$R = \frac{Ip}{P_{in}} \tag{3}$$

So we get:

$$\eta = R\frac{h\nu}{q} \tag{4}$$

Finally:

$$\eta = R\hbar\overline{\omega} \tag{5}$$

This formula is key in evaluating the quantum efficiency of individual solar cells.